Process Management

A process is a program that is currently being executed by the operating system.

Each process owns a private address space that other processes cannot read directly.

The process control block stores the saved registers and the state of a process.

A context switch saves the state of the running process and restores the state of another.

Time spent on a context switch does no useful work for any user program.

A process in the waiting state cannot run until the event it needs has occurred.

CPU Scheduling

Round-robin scheduling gives each process a fixed time quantum on the processor.

A preempted process moves to the back of the ready queue under round-robin scheduling.

Shortest job first scheduling picks the process with the smallest expected burst time.

Priority scheduling can starve low priority processes unless ageing raises their priority.